

New York University — Tandon School of Engineering

Department of Technology Management and Innovation

IE-GY 9113B — Systems Integration: From ERP to Agentic AI

Fall 2026 | Fridays 8:00–10:30 AM | 2 MTC Room 802

September 4 – December 11, 2026

Professor Emanuele Cimica

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Office location and office hours will be published on Brightspace.

Course Prerequisites

Graduate standing. Familiarity with business processes and organisational management is assumed. No prior technical background in ERP systems, automation, or artificial intelligence is required.

Course Description

This is a semester-long course worth 3 credit hours, meeting Fridays 8:00–10:30 AM from September 4 through December 11, 2026 — 14 teaching sessions in total.

This course provides a comprehensive and strategic view of enterprise technology integration, designed to equip the next generation of leaders and technologists with a differentiating capability: the ability to design, integrate, and leverage complex business systems across strategic, operational, and human dimensions.

In today's economy, the Enterprise Resource Planning (ERP) system is not a competitive advantage — it is a table stake. The true competitive advantage lies in what comes next: successful integration and optimisation, unlocking the strategic potential that remains untapped in most organisations' technology backbones. This course treats ERP as the non-negotiable foundation and builds upward through Robotic Process Automation (RPA), Intelligent Automation, and ultimately autonomous AI agents.

The foundational challenge is universal regardless of company size: how do you create a reliable, integrated data backbone from which to build progressive layers of automation? A Fortune 500 company with a \$50 million SAP implementation and a 30-person startup stitching together Salesforce, QuickBooks, and Shopify face identical strategic questions. Where this syllabus refers to 'ERP,' students working in smaller organisations should read 'integration backbone.'

To make this concrete, the course works with four recurring company archetypes: a Consumer Packaged Goods (CPG) manufacturer running S&OP off SAP; a mid-market manufacturer using NetSuite for Order-to-Cash; a Series B startup running order-to-cash across Stripe, Shopify, and QuickBooks; and a 50-person services firm operating on HubSpot and Notion. These four archetypes travel through the full course and give you a portable mental model across every scale of business — from Fortune 500 to seed-stage. Wherever this syllabus refers to 'ERP,' locate your company on this scale and read accordingly.

The business problems this course is designed to address fall into three interrelated threads that run through every module and every lab: **Customer** — how integrated systems enable order fulfilment, service quality, and customer experience at scale; **Operations & Supply Chain** — how production planning, inventory management, and process synchronisation determine whether a company can deliver on its promises; and **Cash Management** — how the speed and reliability of the integration backbone determines how fast cash moves through the business. Cash flow is a leading reason businesses fail. Value growth is a leading reason they succeed. This course teaches you how integration and automation directly move both levers — and how agentic AI is creating the next wave of competitive separation for companies that get this right.

This is not only a technology course. It teaches a **replicable methodology** — problem framing, process diagnosis, solution design, governance, and board-ready communication — that you can deploy professionally from the moment you complete it. The capstone project is structured as a strategic proof of concept: your team picks a real company, identifies the integration and automation opportunity most likely to move the needle on one of the three threads, and builds a board-ready investment case across four progressive labs.

Learn-by-Building: A distinctive feature of this course is its learn-by-building philosophy. Students complete one structured lab per module, using accessible no-code platforms to build and break a process map, an automated workflow, an AI agent, and a governance brief. These four artifacts are progressive — each one builds on the previous — and together they form the evidence base for the final deliverables.

The curriculum is structured around three core pillars present in every module:

- **Foundational Process Integration:** How a reliable data backbone provides the single source of truth and defines the fundamental constraints and opportunities for automation.
- **Strategic Business Integration:** How to layer automation intelligently — from RPA to Agentic AI — on that foundation to create measurable business value.
- **Change Leadership & Ethics:** The managerial and ethical frameworks required to lead people through radical technological change, addressed in every module.

Course Objectives

By the end of this course, you will be able to:

- **Apply a Replicable Methodology:** Use the course framework to diagnose integration and automation opportunities across Customer, Operations & Supply Chain, and Cash Management — and produce board-ready investment cases that quantify value, cost, and risk.
- **Evaluate Strategic Leverage:** Articulate how a well-integrated technology backbone serves as the foundation for the modern enterprise and how to leverage it for continuous strategic enablement.
- **Design Transformation Blueprints:** Design a multi-layered digital transformation roadmap identifying appropriate technologies — from RPA to Agentic AI — layered on the integration foundation.
- **Model and Redesign Processes:** Analyse and redesign core business processes (BPR) optimised for system integration and both rules-based and cognitive automation.
- **Build and Evaluate Agents:** Deploy, test, and critically evaluate AI agents using accessible no-code platforms, identifying failure modes and governance requirements from direct experience.

- **Govern Autonomous Systems:** Examine the ethical, risk, and governance frameworks necessary to safely deploy and manage multi-agent AI systems in enterprise and startup contexts.
- **Lead Organisational Change:** Develop a methodology for diagnosing, leading, and sustaining technology-driven change, recognising that integration success is primarily a function of human and organisational alignment.

Who This Course Is For

This course is open to students across NYU, and supports students in building the following key capabilities:

- **NYU Tandon — Management of Technology (MOT):** Strategic Enablement & Change Leadership. Translate business goals into system requirements, manage organisational resistance to change, and evaluate the ROI of digital investments. Target roles: Product Manager, Program Manager, Technology Consultant, Digital Transformation Manager.
- **NYU Tandon — Industrial, Mechanical, Electrical, Chemical Engineering:** Cyber-Physical Systems & Integrated Automation. Understand how the integration backbone governs physical operations and master the integration of digital AI agents with physical robotics. Target roles: Business Transformation, Continuous Improvement, Operations Research, Operations & Supply Chain Management.
- **NYU Tandon — Computer Engineering:** System Architecture & Autonomous Design. Master how foundational integration layers interface with modern distributed intelligence. Design and govern data pipelines and API layers for intelligent automation. Target roles: Solutions Architect, Automation Engineer.
- **NYU Stern School of Business:** Digital Strategy & Venture Leadership. Build technology literacy grounded in real operational experience. Leave able to evaluate AI transformation investments with technical credibility, lead due diligence on automation initiatives, and design governance frameworks for AI-enabled business models. Target roles: Chief of Staff, Strategy & Operations Leader, Technology Investment Leader, Digital Transformation Leader.

About the Professor

Emanuele Cimica is an Adjunct Professor in the Department of Technology Management and Innovation at NYU Tandon School of Engineering and a global executive leader with over 25 years of experience in business transformation, operational excellence, and digital innovation. He is the Founder and Chief Transformation Architect of NexusNova, a consultancy dedicated to closing the gap between executive intent and operational reality by integrating human capability with advanced AI.

Before founding NexusNova and joining NYU, Emanuele held senior executive roles at Shell, where he served as General Manager of Operated Assets in Brazil — holding full P&L accountability for a \$300M+ operation and a workforce of 600+ — and as General Manager of Business Transformation, leading Shell's internal global transformation consultancy across its highest-value assets.

At NYU Tandon, Professor Cimica's teaching is rooted in Active Learning, focused on equipping students to solve complex, real-world problems at the intersection of technology, operations, and leadership. He serves as Chairman of the Board of the Tripod Foundation and as a Member of the Executive Council for Leading Change (ECLC). He holds an MBA from Henley Management College (UK) and a Master of Chemical Engineering from the University of Rome La Sapienza.

Course Structure

In-Person: This course is conducted entirely in person. All class meetings are held Fridays 8:00–10:30 AM at 2 MTC Room 802. The 2.5-hour (150-minute) session format accommodates both substantive content and active lab work in the same sitting. Lab sessions — one per module — use the full period: approximately 90 minutes of structured build time (60 minutes for Lab 4, which is a synthesis exercise), followed by team checkpoint presentations and a group debrief for the remaining time.

Grade Breakdown

- Participation: 5% - Attendance and participation in class
- Team Engagement: 5 % - Contribution to presentation development and delivery
- Quizzes (2): 20% (10% each)
- Discussion Forums (3): 30% (three forums, 10% each)
- Team assignments: 40% - Final Presentation (20%) and Executive Board Memo (20%)

Grade Scale – Letter grades with +/- : A 94.45 | A- 89.45 | B+ 86.45 | B 82.45 | B- 79.45 | C+ 76.45 | C 72.45 | C- 69.45 | D+ 66.45 | D 62.45 | below 62.45 = F (minimum % for each letter grade).

Learning Time Rubric

Learning Time Element	Time on Task for Students (weekly estimate)	Notes
Weekly Sessions (Lecture or Lab)	2.5 hours	Sessions meet Fridays 8:00–10:30 AM in person. Lab sessions replace lecture approximately once per module.
Study Notes, Readings & Cases	1.5 hours	The weekly study note is the main teaching text and is read before each session. Case readings — and optional textbook chapters for deepening — are assigned by session and available on Brightspace.
Discussion Forum Posts	1 hour	Three forums total. Includes writing your own post and engaging substantively with two peers.
Lab Preparation & Artifact Write-Up	1.5 hours	Four progressive labs. Includes platform setup, in-class build time, and the one-page artifact submitted EOD of each lab week.
Final Project (Team)	2 hours	Progressive team work across all four modules, culminating in a 20-minute presentation (Dec 11) and a 4–5 page Executive Board Memo (Dec 14).

Course Communication

Your professor will communicate through this syllabus, via email, during office hours, and through NYU Brightspace using the following methods:

Announcements

Announcements will be posted on NYU Brightspace to keep you up to date on all course activities and deadlines. It is your responsibility to check regularly.

Discussion Forums

Discussion forums are accessible through the Discussions tab in your NYU Brightspace course site.

Virtual Meetings

In addition to weekly in-person sessions, your professor may hold virtual office hours through the Zoom tool on NYU Brightspace.

Netiquette

Always use professional language in Brightspace posts and be respectful of classmates even when you disagree with their ideas.

Readings

This course has no single mandated textbook. The weekly study note is the primary teaching text; the books below are optional reference works the notes draw on, labelled core, reference, or optional by how central they are to the course. Five of the six titles are held in the NYU Libraries collection; the sixth, Madison (2005), is optional deepening, available via purchase or Interlibrary Loan.

- Westerman, Bonnet & McAfee — **Leading Digital** (HBS Press, 2014) — all modules — **Core text**
- Iansiti & Lakhani — **Competing in the Age of AI** (HBS Press, 2020) — Modules 3–4 — **Core text**
- Tella, Brinker & Pezzini — **The New Automation Mindset** (Wiley, 2023) — Modules 2–3 — **Reference text**
- Bornet, Wirtz et al. — **Agentic Artificial Intelligence** (World Scientific, 2025) — Modules 3–4 — **Reference text**
- Ross, Weill & Robertson — **Enterprise Architecture as Strategy** (HBS Press, 2006) — Module 1 — **Core text**
- Madison — **Process Mapping, Process Improvement and Process Management** (Paton Professional, 2005) — Module 1, optional — **Optional reading**

The weekly study note is the main teaching text — read it before that week's class; it teaches the week's concepts directly. The books listed above are optional reference works you can consult for a fuller treatment, but they are not required reading. Each lecture week's assigned reading is the case study.

Course Requirements

Participation is paramount. Log into NYU Brightspace multiple times a week, complete all readings before each session, and submit lab artifacts on schedule. Lab artifacts are progressive — each one builds on the previous and feeds your final deliverables. A late lab has downstream consequences for your team. All assignment due dates are in the Brightspace calendar.

Discussion Forums — Due Monthly (3 Forums Total)

30% of Total Grade

Three required forum posts across the semester, one per month, aligned with the end of each of Modules 1, 2, and 3. Each post should be approximately 300–350 words and must demonstrate critical analysis of the module's frameworks and case studies — not summary. You must also respond substantively to two peer posts per forum. Substantive engagement means challenging an assumption with evidence, adding a real case or data point, or offering a reasoned alternative — not agreement or summary.

Team Engagement — Peer Evaluation

5% of Total Grade

The Team Engagement mark reflects each member's contribution to the team's work, assessed through a lightweight, confidential peer evaluation: a formative round at the midterm (no grade weight, to surface problems while they are still fixable) and a graded round at the end of term. Each member rates every teammate, and themselves, on a few simple contribution dimensions.

Final Deliverables (Presentation and Executive Board Memo)

40% of Total Grade

The final project is the integrating deliverable of the course. It has two components: a 20-minute team presentation (December 11) and a 4–5 page Executive Board Memo (due December 14). The presentation is the focal point — your team defends its strategy under live questioning. The memo is the professional written document, refined after presenting. Where possible, the final presentations will be evaluated by a panel of external industry practitioners in a competition format; the instructor will confirm the format before the final session.

Both components are built progressively from the four lab artifacts produced across the semester. No separate written report is required. Full project specifications and lab deliverables will be distributed at course launch.

Company Context

You will work in teams of 3–4, composed to mix technical and non-technical backgrounds. You select your company context at the start of the semester, subject to instructor approval by end of Week 1. Your team is encouraged to work with a real organisation — a current employer, a family business, an internship host, a company with which a team member has a meaningful professional relationship, or a company that the team has interest to work on. In any case, all project data is generated synthetically; no confidential company information is used or required. Where possible the project will be sponsored and supported by a NYU industry partner. For each module, your team will identify which of the three core threads — Customer, Operations & Supply Chain, or Cash Management — the module's concepts most directly affect in your chosen company, and frame your lab deliverable accordingly. Team formation, the team charter (a one-page working agreement due end of Week 2), and the peer-evaluation mechanism for the Team Engagement mark are described on Brightspace.

Lab Portfolio — Progressive Artifacts

One lab per module, submitted end-of-day of the lab week. Labs use free, browser-based, no-code platforms and no programming background is assumed. Each artifact maps directly to a section of your final board memo. The four labs map to three board presentation milestones that structure your capstone: **Milestone 1** (after Lab 2) — *What is the problem?* Your diagnosis of the integration gap and where automation adds value. **Milestone 2** (after Lab 3) — *What is the design to close it?* Your agent architecture and governance framework. **Milestone 3** (Lab 4 + Final Presentation) — *How do we manage what we built?* Your three-phase roadmap with value, cost, and risk for each phase.

Technology Stack Flexibility — Choose Your Stack

You meet the intent and deliverables of each lab; how you get there is largely your choice. Each lab specifies a default platform that supports the 90-minute build window and is sufficient for full marks. You are free to meet the lab's intent using an alternative: a partner-supported enterprise path on AWS, or a stack of your own design combining default, AWS, or other platforms. Whatever the stack, the graded standard is identical — business outcomes,

governance thinking, and solution-design quality, never platform choice or complexity. Three paths are open:

- **Default** — the no-code stack: Miro/Lucidchart, Airtable/Sheets, Gemini, Make, Relevance AI. Requires no approval, and is fully sufficient for every grade outcome. A team that wants no tool-selection overhead uses this and is never disadvantaged.
- **Enterprise (AWS) — by approval/availability** - an enterprise-grade path offered through the course's industry collaboration with AWS senior alumni advisor, who support lab environments, credits, and onboarding. Requires pre-lab onboarding and faculty notification.
- **Custom (student-designed) — by approval** — compose your own stack from default, AWS, and/or other platforms. Any non-default stack requires faculty approval and a declared platform choice by end of Week 3; Approval is lightweight and checks only three things: (1) the stack can meet the lab's intent and deliverables; (2) it does not change the graded standard; (3) the tools are acceptable for handling course (synthetic) data.

Assessment fairness: grading is on business outcomes, governance thinking, and solution-design quality — never platform complexity. No stack (default, AWS, or custom) confers a grading advantage; platform choice may be acknowledged only in the participation/engagement component. The default remains fully sufficient for full marks.

Executive Board Memo (20%) — 4–5 Pages — Due December 14

The board memo is your final written deliverable, structured as a professional document a real executive team would present to a board of directors. Submit via Brightspace by end of day December 14. Your memo should reflect the full journey, including insights from the December 11 presentation and from hearing other teams' Q&A. Full structure and section-by-section guidance will be distributed at course launch.

Final Presentation (20%) — December 11 — 20–25 Minutes Per Team

Your team presents its Intelligent Enterprise Blueprint in a board-style format: 20 minutes of presentation followed by 5 minutes of Q&A. All team members are expected to present.

Quizzes + 1:1 Discussion — Week 6 (Midterm) and Week 13 (Final)

20% of Total Grade (10% each)

Two analytical quizzes administered outside the weekly class session via NYU Brightspace with a 48-hour submission window. Expect 400–600 words per question, structured as business arguments — not summaries — citing specific cases and frameworks. Your Midterm covers Modules 1 and 2. Your Final covers Modules 3 and 4 with one integrative question spanning the full course. Closes with a 1:1 discussion with faculty to deepen on quiz answers.

Course Schedule

14 teaching Fridays, 8:00–10:30 AM, September 4 through December 11, 2026. Lab sessions (★) use the full 2.5-hour period for hands-on building. November 27 is Thanksgiving — no class. Weekly readings are assigned by session and available on Brightspace.

MODULE 1: The Integration Foundation | Weeks 1–3 | Sep 4–18

This module establishes the non-negotiable foundation of the modern enterprise: a reliable, integrated technology backbone. We examine how ERPs and SaaS stacks enforce process standardisation, where they fail to deliver on their promise, and how Business Process Re-engineering (BPR) prepares an organisation for automation. The deeper target of BPR is not just eliminating process steps — it is replacing the underlying planning paradigm: most ERPs are still run as batch-push systems where Materials Requirements Planning (MRP) runs overnight, plans cascade downstream, and humans firefight exceptions in the morning. The precondition for automation to deliver real value is shifting toward near-real-time, event-driven operations where decisions happen at the point of variance. The three core threads — Customer, Operations & Supply Chain, and Cash Management — all run through the integration backbone: customer orders flow through Order to Cash (O2C), supply plans run through MRP and Sales and Operations Planning (S&OP), and cash conversion depends on how fast and accurately the backbone processes transactions. Module 1 teaches you to read a company's integration stack as a diagnostic of all three.

Week 0 (Pre-Course Primer): The ERP Floor — before Friday, September 4

- **In brief** — A pre-course primer that sets the ERP floor and the course's tone before Week 1. It covers ERP basics and how ERP types differ, a BPMN / process-mapping primer. This is a study-note primer (notes teach), not mandated textbook reading.
- **Concepts taught** — What an ERP / integration backbone is; the range of ERP types and how their characteristics differ; the shared data model (finance, procurement, inventory, manufacturing, HR); a BPMN / process-mapping primer.
- **Reading set** — Tone-setter: McKinsey, 'Bridging the great AI agent and ERP divide to unlock value at scale' (Jan 2026, open access). Foundation (classic): Davenport, 'Putting the Enterprise into the Enterprise System' (HBR, July–Aug 1998; NYU Libraries) — the classic that defined the problem, not current-state. Supporting video: 'An explanation of SAP ERP, ECC and S/4HANA' (Eye on Tech / Informa TechTarget). Optional: SAP, 'What is ERP: The Essential Guide.'

Week 1: Introduction: The Integration Backbone - Friday, September 4

- **In brief** — The integration backbone — ERP or a multi-SaaS stack — is the single source of truth and the non-negotiable table stake for modern operations at any scale. With the ERP floor set in Week 0, this week defines what the backbone is and why it so often fails to deliver value even when it works technically.
- **Objectives** — Define the backbone as single source of truth and process enforcer; read O2C and P2P as the flows every integration touches; locate control points and decision points; frame backbone work as a business transformation, not an IT project.
- **Concepts taught** — The integration backbone (single source of truth); Order-to-Cash and Procure-to-Pay; control points and decision points; the leverage gap; packaged ERP versus custom-built backbones; the Four Lenses (previewed). (ERP anatomy is covered in the Week 0 primer; the delivery ecosystem and legacy reality are covered in Week 2.)
- **Assigned case** — Primary case: NYU Bobst Library / Jaggaer migration (2022), grounded in Ferguson (2025), ALA LRTS — peer-reviewed, open access. Supporting: Lidl / SAP "eLWIS" (~€500M, abandoned 2018 after ~7 years) , from Computer Weekly and Handelsblatt; the enterprise-scale counterpart to Bobst's small-scale integration gap, showing standardization-vs-local-fit failure.
- **Deliverable → Board Memo** — §2 (Context & Background): confirm your company by end of Week 1 and draft a first backbone-context note

(stack, key processes, archetype, primary thread). You also receive the capstone storyboard this week and map your team's final board deck to it, so each week's homework fills a defined piece of the final presentation.

- **Teams & setup** — You receive your team assignment after this first class; engage with your teammates once teams are formed, and set up free accounts for the default-stack tools (Miro/Lucidchart, Airtable/Google Sheets, Gemini) during the first week.
- **Go deeper (optional)** — Ross & Weill Ch. 1–2 (foundation of execution + operating models); Westerman, Introduction. Optional: Madison on process mapping.

Week 2: Failure to Leverage + BPR as Remedy - Friday, September 11

- **In brief** — A technically 'successful' ERP can still fail to deliver value; the remedy is process redesign before automation, not better technology. Nike shows the failure mode; Hammer's 'obliterate, then automate' and the Four Lenses show the fix.
- **Objectives** — Distinguish leverage failure from implementation failure; sequence BPR ahead of automation; locate value leaks with the Four Lenses; decide where total redesign is warranted and where incremental change suffices; embed adoption from the start of a redesign.
- **Concepts taught** — Leverage failure; Business Process Re-engineering (Hammer); selective redesign — total versus incremental change, concentrating on the few areas where value is created or destroyed rather than reshaping the business to fit the platform; the ERP delivery ecosystem and integration on-ramps; vendor-involvement risk and retained design authority; the legacy / multi-system reality; batch-push vs. event-driven operations; the Four Lenses (Frustration, Time, Cost, Quality); adoption over change management; Lab 1 pre-work (spans W2–3).
- **Assigned case** — Primary case: Birmingham City Council / Oracle (2022–23) — s114 notice + Grant Thornton audit + Sheffield Audit Reform Lab. Supporting: Nike i2 supply-chain failure (2000) — Koch, "Nike Rebounds" (CIO, 2004); the Dolfing blog is a further supporting reference.
- **Deliverable → Board Memo** — §2 (Context & Background): a one-page context-and-diagnosis brief on your company's current backbone and where it fails to leverage. Also due end of Week 2: your team charter (template on Brightspace).

- **Go deeper (optional)** — Ross & Weill develop the operating-model choice in Ch. 2–3; Hammer’s 1990 HBR essay is the origin of ‘obliterate, then automate.’

★ Week 3: LAB 1 — Diagnose & Map a Process - Friday, September 18

- **In brief** — A two-part lab spanning Weeks 2–3. You research a company’s real pain (gated pre-work brief), then in class map its process and control points, absorb a curveball, and generate synthetic data last — business first, data last.
- **Objectives** — Diagnose friction with the Four Lenses; map an end-to-end process, its control points and its decision points; identify the best RPA candidate step and the step that needs AI judgment.
- **Concepts applied** — Friction-chain diagnosis; control points and decision points; the Four Lenses; RPA-candidate vs. AI-judgment steps; rejection and rework paths; the synthetic-data sandbox.
- **Reading** — set up the lab platform before class (accounts are free); the pre-work research brief is due and checked before class.
- **Deliverable → Board Memo** — §3 (Backbone Assessment) — Milestone 1 (problem diagnosis): research brief + annotated process map (control points, three diagnostic points, friction chain) + Gemini-built sandbox.
- **Go deeper (optional)** — set up the lab platform before class.

MODULE 2: The Automation Ladder | Weeks 4–6 | Sep 25–Oct 9

This module builds the automation capability layer: from first-generation RPA through scaled enterprise deployment, governed by a Center of Excellence. Lab 2 produces a working automation scenario that connects at least two systems. The primary thread in Module 2 is Operations & Supply Chain — RPA finds its first traction in repetitive, rules-based operational processes — but every automation decision has a Cash impact (cycle time = cash conversion speed) and a Customer impact (order accuracy = customer experience).

Week 4: RPA: The Tactical Integration Layer - Friday, September 25

- **In brief** — RPA is the first, tactical layer of automation on the backbone. This week defines it, separates it from AI, and shows where it first finds traction — and where its limits appear.

- **Objectives** — Define RPA and distinguish it from traditional automation and AI; identify high-value RPA candidates; avoid automating a broken process; read a working O2C baseline.
- **Concepts taught** — RPA vs. traditional automation vs. AI; the RPA business case and candidate selection; MRP/S&OP batch-push planning; the working-backbone baseline.
- **Assigned case** — Primary case: Thermo Fisher Scientific RPA (CIO.com). Supporting: Bailey Hydraulics / NetSuite O2C, a working-backbone baseline.
- **Deliverable → Board Memo** — §4 (Automation Journey): a short RPA-candidate scan for your company — which repetitive, rules-based steps qualify, and why.
- **Go deeper (optional)** — Tella et al. Ch. 1–2 (Automation + Process Mindset); Deloitte, Automation with intelligence (2022); MRP/S&OP context: Orlicky (1975), Ling & Goddard (1988).

Week 5: Scaling RPA + Center of Excellence - Friday, October 2

- **In brief** — Most RPA programmes stall at a handful of bots. A Center of Excellence governs the jump from pilot to enterprise scale — and defines the human roles that keep automation sustainable.
- **Objectives** — Explain why RPA programmes stall at scale; design a CoE operating model and governance; place the human roles in a scaled programme.
- **Concepts taught** — Scaling limits of RPA; the Center of Excellence (operating model, governance, bot ownership); selector instability and integration drift; building automation alongside the workforce.
- **Assigned case** — Primary case: Telefónica O2 RPA (Lacity & Willcocks, MIS Quarterly Executive, 2016; open-access working paper) — a peer-reviewed account of scaling RPA through a small in-house team and a Back-Office-owned operating model. Supporting: Orange Spain 'Robot Factory' (UiPath case study; vendor-reported), a second telecom Center of Excellence for illustration.
- **Deliverable → Board Memo** — §4 (Automation Journey): a draft CoE / operating-model section for your company.
- **Go deeper (optional)** — Tella et al. Ch. 4–5 (Scale Mindset + Orchestration); Westerman Ch. 3 (Core Operations).

★ Week 6: LAB 2 — Automate the Repeatable Part - Friday, October 9

- **In brief** — Build a working multi-step automation that connects at least two systems for your company — then break it. Finding where it breaks is the governance lesson.

- **Objectives** — Build a multi-step automation in Make; connect two cloud services; document a failure mode and the human oversight it implies.
- **Concepts applied** — Trigger-condition-action workflows; multi-system hand-offs; failure modes and oversight; simple ROI of automation.
- **Reading** — Make.
- **Deliverable → Board Memo** — §4 (Automation Journey) — completes Milestone 1: a working automation + a one-page post-mortem (what broke, oversight, ROI).
- **Go deeper (optional)** — set up Make before class.

MODULE 3: Intelligent Automation + AI Agents | Weeks 7–10 | Oct 16–Nov 6

This module moves from rule-based automation to AI-native systems. We examine the architectural shift from RPA to intelligent automation, the SPAR framework for AI agent design, and the governance challenges of deploying systems that can act autonomously. Lab 3 is the centrepiece: building and deliberately breaking an AI agent. Module 3 is where the Customer thread becomes primary: AI agents are being deployed first at customer-facing interfaces (service, fulfilment, dispute resolution) because that is where the ROI is most visible and the governance stakes are highest.

Week 7: From RPA to Intelligent Automation - Friday, October 16

- **In brief** — Where rule-based automation hits its ceiling and cognitive technologies (NLP, ML) take over — the shift from automating tasks to automating decisions, and what it demands of governance.
- **Objectives** — Identify what rules-based RPA cannot do; explain the shift to event-driven, intelligent automation; articulate the governance change when AI replaces expert judgment.
- **Concepts taught** — Limits of rules-based systems; NLP/ML for unstructured inputs; event-driven vs. batch-push architectures; the Enterprise AI Platform; process intelligence.
- **Assigned case** — Primary case: JPMorgan COiN (Contract Intelligence) — Son, 'JPMorgan Software Does in Seconds What Took Lawyers 360,000 Hours' (Bloomberg, 2017), journalism of record; open-access report via ABA Journal (2017).
- **Deliverable → Board Memo** — §5 (Agent Design & Governance): identify the one step in your process that needs AI judgment (your Lab 3 target) and why rules fail there.

- **Go deeper (optional)** — Tella et al. Ch. 15 (Enterprise AI Platform); Iansiti & Lakhani Ch. 3 (AI Factory); Bornet et al. Ch. 1–2 (pre-read for W8).

Week 8: AI Agent Architecture: The SPAR Framework - Friday, October 23

- **In brief** — The anatomy of an autonomous agent — Sense, Plan, Act, Reflect — and the three keystones (Action, Reasoning, Memory). The Klarna reversal shows what happens when Reflect is missing.
- **Objectives** — Define the SPAR loop and the three keystones; locate the governance requirement at each stage; explain the Act-without-Reflect failure mode.
- **Concepts taught** — SPAR (Sense, Plan, Act, Reflect); the three keystones (Action, Reasoning, Memory); autonomy and the kill-switch question; Reflect as the primary enterprise failure point.
- **Assigned case** — Primary case: Klarna AI customer-service agent (2024–25), from Bloomberg (May 2025). Supporting: CX Dive (Nov 2025) and the Twig.so deep-dive.
- **Deliverable → Board Memo** — §5 (Agent Design & Governance): a one-paragraph SPAR sketch for your Lab 3 agent (goal, tools, where Reflect must sit).
- **Go deeper (optional)** — Bornet et al. Ch. 3 (Inside the Mind of an AI Agent); WEF AI Agents primer (2024).

★ Week 9: LAB 3 — Build an AI Agent - Friday, October 30

- **In brief** — Build a functional agent with a goal, tools, and a SPAR loop — then test it to failure with adversarial inputs. The adversarial test is the governance insight.
- **Objectives** — Build an agent in Relevance AI; design ambiguous, edge, and adversarial inputs; document failures and derive guardrails.
- **Concepts applied** — The SPAR loop in practice; tool use; adversarial testing; guardrails and human-in-the-loop.
- **Reading** — set up the lab platform before class (accounts are free); the Agent Design Brief is due before class.
- **Deliverable → Board Memo** — §5 (Agent Design & Governance) — Milestone 2 (solution design): a shareable agent + a one-page evaluation (three adversarial inputs, results, three guardrails). Governance and ethics: record your human-in-the-loop boundary and your most consequential adversarial failure — you will carry both into the Week 13 Human-AI Partnership argument (Board Memo §7).
- **Go deeper (optional)** — set up Relevance AI before class.

Week 10: Multi-Agent Systems + Mod. 3 Checkpoint - Friday, November 6

- **In brief** — How agents collaborate, delegate, and coordinate — including cyber-physical systems where digital agents govern physical robots. The Module 3 checkpoint reviews your agent design.
- **Objectives** — Describe multi-agent coordination; identify cyber-physical risk categories; present your agent design for structured feedback.
- **Concepts taught** — Multi-agent systems (collaboration, delegation, coordination); cyber-physical systems; coordination mechanisms that avoid deadlock; the backbone as communication layer.
- **Assigned case** — Primary case: Amazon multi-agent warehouse logistics (ResearchGate).
- **Deliverable → Board Memo** — §5 (Agent Design & Governance): the Agent Design Brief + Governance Framework draft (due before class) — the formal Module 3 checkpoint.
- **Go deeper (optional)** — Bornet et al. Ch. 5–6 (Action + Reasoning); Westerman Ch. 4.

MODULE 4: The Agentic Enterprise | Weeks 11–14 | Nov 13–Dec 11

This module zooms out to the enterprise and societal level: how agentic AI restructures industries, what a credible transformation roadmap looks like, and what responsible leadership of human-AI systems requires. Lab 4 synthesises the semester into an Intelligent Enterprise Blueprint. The module closes with final team presentations. Module 4 closes the loop on Cash Management: the board-level argument for any AI investment must show how it moves cash conversion speed, working capital efficiency, or revenue growth — not just cost reduction. Your Blueprint and Board Memo are structured to make that argument

Week 11: Agentic AI Disruption + Strategic Roadmap - Friday, November 13

- **In brief** — How agentic AI disrupts business models, not just processes — and how to build a credible three-phase transformation roadmap the board will fund.
- **Objectives** — Distinguish process optimisation from business-model disruption; build a Stabilise → Automate → Orchestrate roadmap; frame investment, risk, and board communication.

- **Concepts taught** — Business-model disruption vs. process optimisation; compounding automation advantage; the three-phase roadmap (Stabilise, Automate, Orchestrate); board-level framing.
- **Assigned case** — Framing (this forward-looking roadmap week carries perspectives, not an empirical primary case): Retail Banking and agentic AI — a McKinsey perspective (the \$170B figure is McKinsey's own projection, a consultancy viewpoint). Supporting: AWS Forward Deployed Engineering (FDE), the \$1B AWS announcement (June 2026) paired with independent journalism (CNBC and Reuters); it illustrates the shift from advisory to embedded, outcome-based delivery.
- **Deliverable → Board Memo** — §6 (Three-Phase Roadmap) + §1 (Executive Summary): a first draft of your three-phase roadmap with rough value, cost, and risk per phase.
- **Go deeper (optional)** — Westerman Ch. 4–6 (business models, vision, engaging at scale); Iansiti & Lakhani Ch. 6–7; Bornet Ch. 11 (recommended).

★ Week 12 — LAB 4 — Govern What You Built - Friday, November 20

- **In brief** — A synthesis lab — lay your three lab artifacts side by side and produce a one-page governance brief: what would keep you up at night if this ran at scale. The session also runs a brief, formative pulse check on each team's emerging Blueprint and roadmap — feedback, not a grade — so teams can course-correct before the Week 14 final.
- **Objectives** — Identify the highest-risk failure mode across artifacts; specify the oversight point, a monitoring metric, and a shutdown trigger.
- **Concepts applied** — Cross-artifact risk; human oversight points; monitoring and early warning; shutdown conditions; governance-by-policy (platform guardrails, e.g., Bedrock Guardrails) vs. governance-by-design (your own frameworks), which you consolidate in Week 13 (Board Memo §7).
- **Reading** — no new platform; bring your Labs 1–3 artifacts (Miro/Lucidchart for the blueprint).
- **Deliverable → Board Memo** — §6 / governance — Milestone 3 (investment case): a one-page governance brief that becomes a memo section directly.
- **Go deeper (optional)** — set up the lab platform before class.

Week 13: Executive Leadership + Human-AI Partnership - Friday, December 4

- **In brief** — The ethics, governance, and liability of autonomous AI, and the future of work — which roles AI augments, which it displaces, and how organisations build human capability alongside. This week consolidates the governance and ethics threads developed in Labs 3 and 4.
- **Objectives** — Map accountability, explainability, and liability for autonomous AI; describe complementary human-AI teaming; state the governance step that must precede deployment.
- **Concepts taught** — AI accountability and liability; explainability; complementary human-AI teaming; workforce transition and reskilling.
- **Assigned case** — Primary case: Moffatt v. Air Canada (2024 BCCRT 149), from the CanLII ruling and ABA analysis. Supporting: Goldsmith-Pinkham, Tan and Zentefis (2026), 'Human-AI Collaboration in Radiology: The Case of Pulmonary Embolism' (arXiv, open access).
- **Deliverable → Board Memo** — §7 (Human-AI Partnership): draft your workforce-transition and governance-commitments section.
- **Go deeper (optional)** — Borner Ch. 10 & 14; Iansiti & Lakhani Ch. 8; Tella Ch. 20; Westerman Conclusion.

Week 14 — Final Presentations — Intelligent Enterprise Blueprint - Friday, December 11

- **In brief** — Board-style team presentations of the Intelligent Enterprise Blueprint — 20 minutes plus 5 minutes of Q&A, treated as a board deciding whether to fund the roadmap.
- **Objectives** — Defend a coherent problem → design → roadmap argument under live questioning; present as a team.
- **Evaluated on** — clarity of problem framing (Milestone 1); quality of solution design and governance (Milestone 2); credibility of the investment roadmap with value, cost, and risk (Milestone 3).
- **Reading** — none — final presentation day.
- **Deliverable → Board Memo** — the final presentation; the 4–5 page Executive Board Memo is due December 14 (provided on Brightspace).

Course Policies

Generative Artificial Intelligence Use

This course has a deliberately two-sided relationship with AI: you will use it, and you will build with it. The distinction that governs academic integrity here is not which tools you touch, but who authors the design.

AI as a thinking partner (permitted, with disclosure). You may use AI tools — Gemini or any comparable platform — for research assistance, brainstorming, and drafting support. When you do, disclose it in a brief statement at the end of the submission. All analysis, arguments, and conclusions you submit for grading must represent your own thinking.

AI as a builder (your design must be your own). Your lab artifacts must reflect your own design decisions and hands-on platform work. Because each team chooses its own stack — the Default no-code path, the partner-supported AWS path, or a student-designed custom stack — the boundary is defined by authorship of the design, not by a list of tools. Submitting AI-generated workflow configurations or agent architectures — on any platform, whether Make, Relevance AI, AWS Step Functions, Amazon Bedrock, or a tool of your own choosing — without substantial personal design and modification constitutes academic misconduct.

The course itself builds AI agents, so the line is not about avoiding AI; it is about owning the design. Use AI to think and to build — but the architecture, the decisions, and the judgment must be yours, and you must be able to explain and defend them.

Late Work

Lab artifacts are due EOD of the lab week. Because each lab feeds the next, late submission has downstream consequences for your team. Late lab artifacts are accepted up to one week after the deadline with a 10-point deduction. The agent design brief and governance draft (due before Week 10) are hard deadlines — the checkpoint cannot proceed without them. The final presentation (December 11) and board memo (December 14) are absolute deadlines; no late submissions accepted.

Moses Center Statement of Disability

If you are a student with a disability who is requesting accommodations, please contact New York University's Moses Center for Student Accessibility (CSA) at 212-998-4980 or mosescsa@nyu.edu. You must be registered with CSA to receive accommodations. Information can be found at www.nyu.edu/csa. The Moses Center is located at 726 Broadway on the 2nd floor. If you are anticipating accommodations, please communicate with the instructor no later than end of Week 2.

NYU School of Engineering Policies and Procedures on Academic Misconduct

(From the School of Engineering Student Code of Conduct)

- Introduction: The School of Engineering encourages academic excellence in an environment that promotes honesty, integrity, and fairness. Any act of academic dishonesty is seen as an attack upon the School and will not be tolerated. Students are responsible for familiarising themselves with the School's Policy on Academic Misconduct.
- Definition: Academic dishonesty may include misrepresentation, deception, dishonesty, or any act of falsification committed by a student to influence a grade or other academic evaluation. Academic dishonesty also includes intentionally damaging the academic work of others or assisting other students in acts of dishonesty. Common examples include:
 - Cheating: intentionally using or attempting to use unauthorised notes, books, electronic media, or electronic communications in an exam; talking with fellow students or looking at another person's work during an exam; submitting work prepared in advance for an in-class examination.
 - Fabrication: including but not limited to, falsifying experimental data and/or citations.
 - Plagiarism: intentionally or knowingly representing the words or ideas of another as one's own in any academic exercise; failure to attribute direct quotations, paraphrases, or borrowed facts or information.
 - Unauthorised collaboration: working together on work that was meant to be done individually.
 - Duplicating work: presenting for grading the same work for more than one project or in more than one class, unless express and prior permission has been received from the course instructor(s) or research adviser involved.
 - Forgery: altering any academic document, including, but not limited to, academic records, admissions materials, or medical excuses.

Access the entire School of Engineering Student Code of Conduct at:
engineering.nyu.edu/academics/code-of-conduct

Statement on Inclusion

The NYU Tandon School values an inclusive and equitable environment for all our students. I hope to foster a sense of community in this class and consider it a place where individuals of all backgrounds, beliefs, ethnicities, national origins, gender identities, sexual orientations, religious and political affiliations, and abilities will be treated with respect. If this standard is not being upheld, please feel free to speak with me.